

AP-E4: A Physical Tangent Fermion from Vortex Moduli-Space SQM

Canonical Spin^c Quantization, Conditional $\mathcal{O}(2)$ Twisting, Chirality, Partner Spectrum, Gap, and the AP-E3 Composability Gate

Route F research note

16 July 2026

Abstract

This note supplies the missing intrinsic tangent-fermion realization behind AP-E4, while showing that the vacuum-line polarization remains independent physical data. A half-BPS non-Abelian $U(2)$ Chern–Simons–Higgs vortex has a normalizable internal moduli space $SU(2)_{c+f}/U(1) \simeq \mathbb{CP}^1$. Promoting its orientation to a slow worldline coordinate yields the exact Fubini–Study kinetic term

$$L_B = C \frac{|\dot{w}|^2}{(1 + |w|^2)^2}, \quad C = \frac{2\pi}{e^2} I_0 > 0,$$

where I_0 is an explicitly convergent radial L^2 integral. The one complex unbroken supercharge supplies a complex fermion satisfying $\psi^v = -w^{-2}\psi^w$ under $v = 1/w$, so the physical worldline fermion is intrinsically tangent-valued. This closes the “bosonic fluctuation is not a fermion” objection for this mother model.

The decisive distinction is then made explicit. In canonical Spin^c /Dolbeault polarization the tangent fermion generates the Clifford module $\Omega^{0,*}(\mathbb{CP}^1)$; it does *not* also make the wavefunction $T\mathbb{CP}^1 = \mathcal{O}(2)$ -valued. Consequently the declared canonical Spin^c re-quantization of the untwisted complex has one Dolbeault ground state, not three; the source-selected half-form/operator ordering instead has no normalizable zero mode. An independent coefficient line $E = \mathcal{O}(2)$ changes the supercharge to $Q_E = \sqrt{2} \bar{\partial}_E$ and gives $(h^0, h^1) = (3, 0)$. Its exact nonzero tower is

$$\lambda_{n,\pm} = \pm \frac{2}{\sqrt{C}} \sqrt{n(n+3)}, \quad \text{mult}(\lambda_{n,\pm}) = 2n+3, \quad n \geq 1,$$

so $\Delta_D = 4/\sqrt{C}$ and $\Delta_H = 8/C$ for $H = D^2/2$.

AP-E3’s oriented level $k = +2$ mixed-WZW term could provide precisely this coefficient line, but only after proving that its charged-two-colour $B = 1$ soliton has the *same* $\mathbb{CP}^1$ collective-coordinate map and that the spatially transgressed WZW line pulls back with Chern number $+2$. The vortex used here is an independent $2 + 1$ -dimensional mother model, not that soliton. Thus `physical_tangent_fermion_in_intrinsic_sqm_derived=true`, while the charged-two-colour embedding, AP-E4 worldline completion, Route-E portal, and physics promotion remain false. Product compactification is retained only as a backup behind a full six-dimensional anomaly gate.

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1 Problem and theorem-level answer

The AP-E4 question contains four logically distinct layers:

- (E4.1) derive a physical $\mathbb{CP}^1$ collective-coordinate manifold and its L^2 metric from a mother action;
- (E4.2) derive a Grassmann-odd tangent partner and the worldline supercharge;
- (E4.3) determine which additional line bundle, if any, twists the quantum states and then solve chirality, partners, and the gap;
- (E4.4) prove that the result composes with AP-E3 and eventually Route E.

The first two layers are solved by the intrinsic vortex model below. The third is solved mathematically both without and with a declared $\mathcal{O}(2)$ coefficient line. The fourth remains a sharply stated physical gate.

Theorem 1.1 (Intrinsic tangent result and fixed-polarization line-twist result). *For a charge-one half-BPS $U(2)$ non-Abelian Chern–Simons vortex with the translational coordinate factored out, the internal moduli are $\mathbb{CP}^1$, their L^2 metric is Fubini–Study with a finite positive coefficient C , and the BPS collective-coordinate supersymmetry supplies a physical tangent fermion. In the declared canonical Dolbeault re-quantization the untwisted ground-state space is $H^{0,*}(\mathbb{CP}^1, \mathcal{O}) = \mathbb{C}$. Within $SU(2)$ -equivariant holomorphic line twists of this fixed canonical operator, and with no added superpotential or extra Fermi multiplet, exactly three positive-chirality ground states occur if and only if the independent coefficient line is $E = \mathcal{O}(2)$:*

$$H^{0,*}(\mathbb{CP}^1, E) = H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq \mathbb{C}^3, \quad H^1(\mathbb{CP}^1, \mathcal{O}(2)) = 0.$$

Every nonzero level is paired. This theorem does not claim uniqueness among all possible SQM field contents, potentials, or vacuum-line regularizations.

2 A BPS mother model with internal $\mathbb{CP}^1$

2.1 Bulk ingredients and colour–flavour locking

Consider the 2 + 1-dimensional $\mathcal{N} = 2$ supersymmetric Chern–Simons–Higgs theory with gauge group $U(2)$, two fundamental Higgs flavours assembled into a 2×2 matrix H , and a positive FI scale ξ . Its bosonic action has the schematic but sufficient form

$$S_{\text{bulk}} = \int d^3x \left[\frac{\kappa}{4\pi} \epsilon^{\mu\nu\rho} \text{Tr} \left(A_\mu \partial_\nu A_\rho - \frac{2ie}{3} A_\mu A_\nu A_\rho \right) + \text{Tr}(D_\mu H)^\dagger D^\mu H - V_6(H; \xi, \kappa, e) + \mathcal{L}_F \right]. \quad (1)$$

Supersymmetry fixes the sextic potential V_6 and admits first-order BPS vortex equations. We use the explicit BPS family and collective-coordinate reduction of Ref. [1]; no claim below depends on inventing a new radial solution.

The asymmetric vacuum may be chosen as

$$H_\infty = \sqrt{\xi} \mathbf{1}_2. \quad (2)$$

Gauge $SU(2)_c$ and flavour $SU(2)_f$ are then locked to a global diagonal $SU(2)_{c+f}$. A reference unit vortex embeds its winding in one colour–flavour direction and has stabilizer $U(1)$. Therefore its internal orbit is

$$\mathcal{M}_{\text{int}} = \frac{SU(2)_{c+f}}{U(1)} \simeq \mathbb{CP}^1. \quad (3)$$

The full one-vortex moduli space contains also its centre $Z \in \mathbb{C}$: $\mathcal{M}_1 \simeq \mathbb{C}_Z \times \mathbb{CP}^1$. We factor out Z , or equivalently hold the vortex centre fixed with a weak supersymmetric trap. This avoids confusing translational states with the proposed three-state internal sector.

2.2 Projector representation and physical zero mode

Write a normalized orientation and its rank-one projector as

$$n \in \mathbb{C}^2, \quad n^\dagger n = 1, \quad n \sim e^{i\alpha} n, \quad P = nn^\dagger. \quad (4)$$

The oriented vortex fields can be expressed schematically as

$$H(r; n) = \sqrt{\xi} \mathbf{1}_2 + (\tilde{\varphi}(r) - \sqrt{\xi}) P, \quad A_\mu(r; n) = A_\mu^{(0)}(r) \mathbf{1}_2 + A_\mu^{(1)}(r) \left(P - \frac{1}{2} \mathbf{1}_2 \right). \quad (5)$$

An infinitesimal global rotation generated by a Hermitian $\epsilon \in \mathfrak{su}(2)$ gives

$$\delta_\epsilon P = i[\epsilon, P], \quad \delta_\epsilon H = i[\epsilon, H], \quad \delta_\epsilon A_\mu = i[\epsilon, A_\mu]. \quad (6)$$

Differentiating a family of exact solutions along an exact symmetry orbit annihilates the linearized field equations. Directions satisfying $[\epsilon, P] = 0$ are the stabilizer and hence zero; the remaining two real directions span $T_P \mathbb{CP}^1$.

Normalizability follows from the local-vortex boundary conditions. Near the core all profiles are regular, so the radial norm has a bounded integrand times $r dr$. At infinity the massive bulk equations imply $\tilde{\varphi} - \sqrt{\xi} = O(e^{-mr})$ and analogous decay for the non-Abelian gauge profile. Hence

$$\|\delta_\epsilon \Phi\|_{L^2}^2 \leq C_0 \int_0^1 r dr + C_\infty \int_1^\infty r e^{-2mr} dr < \infty. \quad (7)$$

This is a genuine normalizable collective mode, not merely a tangent vector in field-value space. Index and moduli-matrix analyses of non-Abelian vortices give the same moduli count in broader Yang–Mills–Chern–Simons settings [2].

3 First-principles L^2 metric

3.1 Horizontal geometry of the normalized doublet

For a path $n(t)$, the physical horizontal velocity is

$$\eta = (1 - P)\dot{n}, \quad n^\dagger \eta = 0, \quad \|\eta\|^2 = \dot{n}^\dagger \dot{n} - |n^\dagger \dot{n}|^2. \quad (8)$$

Because $n^\dagger n = 1$, $n^\dagger \dot{n}$ is purely imaginary and therefore

$$\dot{n}^\dagger \dot{n} + (n^\dagger \dot{n})^2 = \dot{n}^\dagger \dot{n} - |n^\dagger \dot{n}|^2 = \|\eta\|^2. \quad (9)$$

On the north chart choose

$$n_N(w) = \frac{(1, w)^T}{\sqrt{1 + |w|^2}}. \quad (10)$$

Direct differentiation and projection give

$$\|(1 - P)\dot{n}_N\|^2 = \frac{|\dot{w}|^2}{(1 + |w|^2)^2}. \quad (11)$$

This proves the Fubini–Study shape without invoking symmetry uniqueness.

3.2 Radial norm and elimination of the temporal compensator

Promoting n to $n(t)$ requires a temporal gauge-field compensator with a radial function $\rho(r)$. Integrating the explicit bulk ansatz over the transverse plane gives [1]

$$S_{\text{eff}} = \frac{2\pi}{e^2} I_0 \int dt \left[\dot{n}^\dagger \dot{n} + (n^\dagger \dot{n})^2 \right], \quad (12)$$

$$I_0 = \int_0^\infty d\hat{r} \hat{r} \left[(\hat{\varphi}^2 + \hat{\xi})\rho^2 + 2(1 - \rho)(\hat{\varphi} - \sqrt{\hat{\xi}})^2 \right], \quad (13)$$

$$\rho = \frac{(\hat{\varphi} - \sqrt{\hat{\xi}})^2}{\hat{\varphi}^2 + \hat{\xi}}. \quad (14)$$

Here hats denote the dimensionless radial variables used in that reduction. The last equation is algebraic because ρ is nondynamical in the pure Chern–Simons model. Substitution makes finiteness manifest from the same core regularity and exponential tail used in Eq. (7); the integrand is nonnegative and nonzero for a nontrivial vortex, so

$$C := \frac{2\pi}{e^2} I_0 > 0. \quad (15)$$

Combining Eqs. (9)–(15) yields

$$\boxed{L_B = C \frac{\dot{w}\dot{\bar{w}}}{(1 + |w|^2)^2}}, \quad K = C \log(1 + |w|^2), \quad g_{w\bar{w}} = \frac{C}{(1 + |w|^2)^2}. \quad (16)$$

Since a round sphere of radius R has $ds^2 = 4R^2 dw d\bar{w} / (1 + |w|^2)^2$,

$$R = \frac{\sqrt{C}}{2}, \quad \text{Area}(\mathbb{CP}^1) = \pi C. \quad (17)$$

For deterministic regression, the companion code also inserts the analytic surrogate $\hat{\varphi} = \tanh \hat{r}$, $\hat{\xi} = 1$ into Eqs. (13)–(14), and verifies positivity and quadrature convergence. This is a numerical check of the reduction, not a claim to have solved the microscopic BPS radial equation.

Fail-closed gate 3.1 (No hidden level-two term in this mother reduction). For the particular ansatz of Ref. [1], the Chern–Simons density has no orientation-dependent term linear in $\dot{n}$; all time-dependent orientational terms in Eq. (12) come from the scalar kinetic energy. Thus this mother model derives C but does not derive a magnetic line of Chern number two. Other Chern–Simons vortex theories can induce magnetic fields on moduli space [3]; that possibility does not retroactively insert one here.

4 Physical tangent fermion and intrinsic $N = 2$ SQM

4.1 BPS supersymmetry and tangent covariance

The bulk vortex preserves one complex supercharge. Supersymmetry therefore completes Eq. (16) with a complex Grassmann collective coordinate $\psi^w(t)$:

$$L_{\text{SQM}} = g_{w\bar{w}} \dot{w} \dot{\bar{w}} + \frac{i}{4} g_{w\bar{w}} (\bar{\psi}^{\bar{w}} D_t \psi^w - \bar{D}_t \bar{\psi}^{\bar{w}} \psi^w), \quad (18)$$

with

$$D_t \psi^w = \dot{\psi}^w + \Gamma_{ww}^w \dot{w} \psi^w, \quad \Gamma_{ww}^w = g^{w\bar{w}} \partial_w g_{w\bar{w}} = -\frac{2\bar{w}}{1 + |w|^2}. \quad (19)$$

Under the south-chart coordinate $v = 1/w$,

$$\psi^v = \frac{\partial v}{\partial w} \psi^w = -w^{-2} \psi^w, \quad D_t \psi^v = -w^{-2} D_t \psi^w. \quad (20)$$

Hence $\psi \in \Gamma(T^{1,0}\mathbb{CP}^1)$ and its kinetic norm is chart independent. This is the physical tangent fermion missing from a purely bosonic AP-E4 argument.

4.2 What the tangent fermion quantizes to

Fix the canonical Spin^c polarization associated with the complex structure. The tangent fermion and its adjoint become one Clifford creation–annihilation pair. Via the metric, a creation operator wedges by a $(0, 1)$ -form and its adjoint contracts. Therefore the Fock bundle is

$$W_{\text{can}} = \Omega^{0,0}(\mathbb{CP}^1) \oplus \Omega^{0,1}(\mathbb{CP}^1) = \Omega^{0,*}(\mathbb{CP}^1), \quad \Gamma = (-1)^q \quad \text{on } \Omega^{0,q}. \quad (21)$$

The supercharges and Hamiltonian are

$$Q = \sqrt{2} \bar{\partial}, \quad Q^\dagger = \sqrt{2} \bar{\partial}^\dagger, \quad D = Q + Q^\dagger, \quad H = \frac{1}{2} \{Q, Q^\dagger\} = \frac{1}{2} D^2. \quad (22)$$

In particular $Q^2 = (Q^\dagger)^2 = 0$, $\{D, \Gamma\} = 0$, and supersymmetric ground states are Dolbeault cohomology classes.

The phrase “tangent fermion” describes the transformation of the *Clifford generator*. It does not say that the Fock vacuum or every wavefunction carries an additional factor $T^{1,0}\mathbb{CP}^1$. Declaring

$$\Omega^{0,*} \longrightarrow \Omega^{0,*} \otimes T^{1,0}\mathbb{CP}^1 \quad (23)$$

is an independent twist, not a consequence of Eq. (20).

Remark 4.1 (Vacuum-line ordering). The explicit operator ordering used in the original vortex quantization packages the two Fock components with a half-form factor and reports no normalizable zero mode [1]; geometrically this is the ordinary-spin realization $\Omega^{0,*} \otimes K^{1/2}$, with $K^{1/2} = \mathcal{O}(-1)$ on S^2 . AP-E4’s operator card instead declares the canonical Spin^c realization (21). The physical tangent fermion is intrinsic, whereas the vacuum-line regularization is an additional datum that must remain fixed when comparing spectra. In particular, the mother action does not derive the canonical vacuum line used in the subsequent spectral theorem. Neither choice supplies the desired coefficient $\mathcal{O}(2)$.

5 Untwisted canonical negative controls and the independent $\mathcal{O}(2)$ twist

5.1 The untwisted model does not have three vacua

For the canonical realization, Hodge theory gives

$$\text{Ker } D^+ \simeq H^0(\mathbb{CP}^1, \mathcal{O}) = \mathbb{C}, \quad \text{Ker } D^- \simeq H^1(\mathbb{CP}^1, \mathcal{O}) = 0. \quad (24)$$

Thus the declared canonical Spin^c re-quantization has one ground state. As stressed in the vacuum-line remark, this is not the source-selected half-form ordering, which has no normalizable zero mode. A different model with enough fermions to realize the full de Rham complex has $b_0 + b_2 = 2$ vacua on S^2 . This also is not a three-state mechanism.

5.2 Magnetic/Wess–Zumino coefficient line

Let $E = \mathcal{O}(k)$ be an independently derived Hermitian line. In north and south patches choose

$$a_N^{(k)} = \frac{k}{2}(1 - \cos \theta) d\phi, \quad a_S^{(k)} = -\frac{k}{2}(1 + \cos \theta) d\phi, \quad (25)$$

$$F^{(k)} = da^{(k)} = \frac{k}{2} \sin \theta d\theta \wedge d\phi, \quad \frac{1}{2\pi} \int_{S^2} F^{(k)} = k. \quad (26)$$

The overlap transition has winding k , so integrality is necessary for a global quantum line. A worldline term

$$S_{\text{WZ}} = \hbar \int a_i^{(k)}(q) \dot{q}^i dt \quad (27)$$

and its supersymmetric curvature coupling replace Eq. (22) by

$$\mathcal{H}_E = L^2 \Omega^{0,*}(\mathbb{CP}^1, E), \quad Q_E = \sqrt{2} \bar{\partial}_E, \quad D_E = \sqrt{2}(\bar{\partial}_E + \bar{\partial}_E^\dagger), \quad H_E = \frac{1}{2} D_E^2. \quad (28)$$

For all integer k , Riemann–Roch and Serre duality give

$$h^0(\mathcal{O}(k)) = \max(k + 1, 0), \quad h^1(\mathcal{O}(k)) = \max(-k - 1, 0), \quad (29)$$

$$\text{ind } D_E^+ = h^0 - h^1 = k + 1. \quad (30)$$

At positive level two,

$$\boxed{\dim \text{Ker } D_{\mathcal{O}(2)}^+ = 3, \quad \dim \text{Ker } D_{\mathcal{O}(2)}^- = 0, \quad \text{ind } D_{\mathcal{O}(2)}^+ = 3.} \quad (31)$$

A homogeneous-coordinate basis is $\{z_0^2, z_0 z_1, z_1^2\}$. The canonical Spin^c determinant after twisting is

$$\det W_E = K^{-1} \otimes E^2 = \mathcal{O}(2k + 2), \quad k = 2 : \quad \det W_E = \mathcal{O}(6). \quad (32)$$

Orientation reversal is shifted in canonical Spin^c convention: three negative-chirality modes require $E = \mathcal{O}(-4)$, for which $(h^0, h^1) = (0, 3)$, not $E = \mathcal{O}(-2)$.

Fail-closed gate 5.1 (CPT and antiparticle polarization). AP-E3 assigns $k = -2$ to the $B = -1$ antibaryon. Holding the same canonical complex polarization fixed would give $(h^0(\mathcal{O}(-2)), h^1(\mathcal{O}(-2))) = (0, 1)$, not the CPT conjugate of the three-state $B = +1$ kernel. A physical completion must therefore derive how CPT conjugates the complex/ Spin^c vacuum line—equivalently, why the antiparticle operator is the anti-canonical conjugate, or why its effective fixed-polarization coefficient is $\mathcal{O}(-4)$. This polarization map is not fixed by the signed WZW integer alone and remains an independent composability gate.

6 Exact partner spectrum and spectral gap

For $k \geq 0$, the monopole-harmonic multiplet at excitation number n has

$$j = \frac{k}{2} + n, \quad \dim V_j = k + 2n + 1. \quad (33)$$

The connection Laplacian and the Kähler–Weitzenböck curvature shift combine to

$$D_E^2 = \frac{1}{R^2} \left[j(j+1) - \frac{k^2}{4} - \frac{k}{2} \right] = \frac{n(n+k+1)}{R^2} = \frac{4n(n+k+1)}{C}. \quad (34)$$

At $n = 0$ this reproduces $k + 1$ positive-chirality zero modes. For $n \geq 1$, oddness of D_E pairs the two chiral sectors and gives the full spectrum [4]

$$\lambda_{n,\pm} = \pm \frac{2}{\sqrt{C}} \sqrt{n(n+k+1)}, \quad \text{mult}(\lambda_{n,\pm}) = k + 2n + 1. \quad (35)$$

For $k = 2$,

$$\Delta_D = \frac{4}{\sqrt{C}}, \quad \Delta_H = \frac{1}{2} \Delta_D^2 = \frac{8}{C}, \quad \text{mult}(\pm \Delta_D) = 5. \quad (36)$$

Every massive state has a partner. Only the kernel carries a chiral imbalance; therefore “no partner” must never be applied to the full tower.

7 AP-E3/AP-E4 composability: necessary and sufficient data

AP-E3 has produced an anomaly-consistent intermediate mixed-WZW candidate whose signed collective term is intended to have $k = n_c B = +2$. That datum can supply the line $E = \mathcal{O}(2)$ in Eq. (28), but the numerical equality of two integers is not enough.

Let Σ_3 be a spatial slice carrying the unit soliton, $\tilde{\mathcal{C}}_{B=1}$ the corresponding framed field-configuration component before quotienting gauge transformations, and

$$\text{ev} : \Sigma_3 \times \tilde{\mathcal{C}}_{B=1} \longrightarrow X \quad (37)$$

the evaluation map into the mixed-WZW target. For the AP-E3 character $\hat{\Omega}_5 \in \hat{H}^5(X; \mathbb{Z})$, fiber integration in differential cohomology first transgresses the degree-five WZW character to the degree-two differential line class

$$\hat{\mathcal{L}}_{\text{WZW}} := \int_{\Sigma_3} \text{ev}^* \hat{\Omega}_5 \in \hat{H}^2(\tilde{\mathcal{C}}_{B=1}; \mathbb{Z}). \quad (38)$$

This degree-lowering step is mandatory: a degree-five class cannot be directly pulled back and equated to a first Chern class. Put $\mathcal{C}_{B=1} := \tilde{\mathcal{C}}_{B=1}/\mathcal{G}$. Gauge consistency further requires the transgressed differential line to be basic and descend:

$$\hat{\mathcal{L}}_{\text{WZW}} = p^* \hat{\mathcal{L}}_{\text{WZW}}^{\text{quot}}, \quad p : \tilde{\mathcal{C}}_{B=1} \rightarrow \mathcal{C}_{B=1}. \quad (39)$$

Failure of this descent is precisely a global gauge-anomaly obstruction.

Let $\mathcal{M}_{B=1}$ be the collective-coordinate component of the charged-two-colour soliton and let

$$\iota : \mathcal{M}_{B=1} \longrightarrow \mathcal{C}_{B=1} \quad (40)$$

embed it into the UV configuration space modulo gauge transformations. The AP-E3 and AP-E4 cards are composable only if all of the following are proved:

(C1) $\mathcal{M}_{B=1} \simeq \mathbb{CP}^1$ with a finite L^2 metric and a positive gap to every noncollective fluctuation;

- (C2) the projector $P = nn^\dagger$ is the same physical orientation in both constructions, rather than an abstractly isomorphic sphere;
- (C3) $B[\iota(n)] = +1$ is constant on this component and the orientation map has degree +1;
- (C4) Eq. (39) holds, including every large gauge transformation, so the transgressed line exists on the same quotient that contains the collective moduli;
- (C5) the transgressed differential WZW line pulls back as

$$\iota^* \widehat{\mathcal{L}}_{\text{WZW}}^{\text{quot}} = 2 \widehat{c}_1(\mathcal{O}(1)), \quad \frac{1}{2\pi} \int_{\mathcal{M}_{B=1}} \iota^* F_{\text{WZW}} = +2; \quad (41)$$

- (C6) the pullback admits the supersymmetric completion in Eq. (28) and introduces no extra massless Fermi or gauge modes;
- (C7) CPT maps the $B = +1$ canonical polarization to the correct anti-canonical $B = -1$ operator, so the antibaryon has the conjugate three-state kernel rather than the one-state fixed-polarization $\mathcal{O}(-2)$ kernel;
- (C8) the collective states carry the required microscopic representation, survive all gauge/anomaly projections, and couple through a Route-E portal below both the bulk and SQM gaps.

When (C1)–(C8) hold, the composition is a theorem:

$$k_{\text{eff}} = n_c B \deg(\iota) = 2 \cdot 1 \cdot 1 = +2, \quad E_{\text{eff}} = \mathcal{O}(+2), \quad (42)$$

and Eq. (31) follows. If the degree is -1 , or if the WZW line lives on a different sphere, the conclusion does not follow.

Fail-closed gate 7.1 (Mother-model mismatch). The mother model in Sections 2–4 is a $2 + 1$ -dimensional supersymmetric $U(2)$ Chern–Simons–Higgs vortex. It independently proves that a physical tangent fermion can arise on $\mathbb{CP}^1$, but it is not the current four-dimensional charged-two-colour $SU(2)_c \times U(1)_g$ AP-E3 candidate and does not prove Eq. (40). Therefore both the charged-two-colour embedding flag and the AP-E4 worldline-completion flag remain false.

Three microscopic ways to close the coefficient-line gate should be pursued in parallel:

1. derive Eq. (41) directly from the AP-E3 mixed-WZW differential character on an explicit stable $B = 1$ solution;
2. calculate a bulk Callias/Fermi-zero-mode index over the soliton and show that its determinant line over $\mathcal{M}_{B=1}$ is $\mathcal{O}(+2)$;
3. derive an exactly-two gapped microscopic doublet whose Berry determinant over the *same* moduli sphere has Chern number +2, and prove that integrating it out preserves supersymmetry and the gap.

8 Backup product compactification and its six-dimensional anomaly gate

A product construction $M^{3,1} \times \mathbb{CP}^1$ could instead turn the internal operator into a four-dimensional chiral multiplicity via

$$\not{D}_6 = \not{D}_4 \otimes 1 + \gamma_5 \otimes D_E, \quad \Gamma_7 = \gamma_5 \otimes \Gamma_2. \quad (43)$$

This route is not used in the present proof. It is admissible only after a six-dimensional chiral field content passes all of the following:

- (6D1) the complete local eight-form anomaly polynomial, including $\text{Tr } F^4$, $(\text{Tr } F^2)^2$, $p_1(T) \text{Tr } F^2$, $p_1(T)^2$, and $p_2(T)$, vanishes or factorizes as $I_8 = \frac{1}{2} \Omega_{\alpha\beta} X_4^\alpha X_4^\beta$ with an integral Green–Schwarz lattice [5];
- (6D2) global gauge/gravitational anomalies are checked by the appropriate spin/ Spin^c bordism invariant rather than inferred from I_8 ;
- (6D3) the Spin^c determinant and $\mathcal{O}(2)$ flux obey integral and mod-two quantization conditions;
- (6D4) the index-weighted four-dimensional spectrum cancels every local and global anomaly, and the KK tower remains above the portal scale.

None is established here, so `six_dimensional_anomaly_gate_closed=false`.

9 Deterministic verification

The companion script `route_f/code/verify_ap_e4_moduli_space_sqm.py` uses a fixed seed and contains no fitted spectral data. It verifies:

1. horizontal projection, the two-chart Fubini–Study metric, tangent fermion covariance, and invariant fermion norm at 250 deterministic complex probes;
2. positivity and convergence of the eliminated- ρ analytic radial surrogate;
3. Riemann–Roch and Serre duality for $\mathcal{O}(k)$, $-6 \leq k \leq 6$, including the untwisted, $k = 2$, $k = -4$, and de Rham controls;
4. midpoint curvature quadrature and transition-function winding for $c_1(\mathcal{O}(2)) = 2$;
5. a finite exact superalgebra block testing $Q^2 = 0$, Hermiticity, $\{D, \Gamma\} = 0$, $D^2 = 2H$, three positive-chirality zero modes, the first gap, and five states per massive sign;
6. fail-closed machine-readable fields separating intrinsic tangent physics from the charged-two-colour embedding, WZW pullback, Route-E portal, and six-dimensional anomaly gate.

It writes

```
route_f/output/ap_e4_moduli_space_sqm.json
route_f/output/ap_e4_moduli_space_sqm.md.
```

A green run certifies the declared geometry and algebra; it deliberately does not turn an open embedding into a physical theorem.

10 Claim ledger and ordered next steps

Claim	Status	Boundary
Independent BPS mother has internal $\mathbb{CP}^1$ and finite L^2 metric	derived	published radial reduction plus chart proof
Physical tangent fermion in intrinsic SQM	derived	BPS collective-coordinate supersymmetry and Eq. (20)
Canonical Spin^c /Dolbeault algebra and spectrum	solved	declared canonical vacuum-line polarization

Claim	Status	Boundary
Mother model selects that canonical vacuum line	false	source-selected half-form ordering has no
Untwisted tangent SQM gives three vacua	false	normalizable zero mode one vacuum only in the declared canonical re-quantization; source ordering has none and the de Rham control has two
Independent $\mathcal{O}(+2)$ coefficient gives three positive zero modes	proved	Riemann–Roch, Hodge theory, and spectrum
This mother model itself derives $\mathcal{O}(+2)$	conditionally false	no orientation-dependent first-order term in its reduction
Same charged-two-colour $B = 1$ $\mathbb{CP}^1$ moduli map	open	no stable AP-E3 soliton/embedding theorem
Signed transgressed AP-E3 WZW line is $\mathcal{O}(+2)$ on that same moduli	open	Eq. (41) not computed
CPT maps the $B = -1$ sector to the conjugate three-state operator	open	fixed canonical $E = \mathcal{O}(-2)$ has only one negative mode
Six-dimensional product anomaly gate	open backup	product route unused; I_8 , GS, and bordism audits open
Three states are Route-E chiral families	open	representation, portal, anomaly, and scale gates unresolved

The ordered research programme is:

1. establish or rule out a stable charged-two-colour $B = 1$ soliton with $\mathbb{CP}^1$ internal moduli and compute its full fluctuation gap;
2. transgress the signed mixed-WZW differential character along the spatial three-cycle, pull the resulting differential line to that exact family, and derive its supersymmetric worldline completion;
3. independently compute the soliton Fermi/Callias index and determinant line, providing a non-WZW cross-check of $\mathcal{O}(+2)$;
4. derive the vacuum-line regularization and its CPT/anti-canonical map rather than imposing the canonical polarization only in the baryon sector;
5. only after those agree, construct the Route-E portal below $\min(\Delta_{\text{bulk}}, 4/\sqrt{C})$ and redo the anomaly ledger;
6. retain product compactification only if its complete 6D local and global anomaly gate can be closed.

The final fail-closed status is

$$\begin{aligned}
&\text{physical_tangent_fermion_in_intrinsic_sqm_derived}=\text{true}, \\
&\text{mother_model_vacuum_line_matches_canonical_spinc}=\text{false}, \\
&\quad \text{o2_twist_three_positive_zero_modes}=\text{true}, \\
&\quad \text{cpt_antibaryon_polarization_closed}=\text{false}, \\
&\text{charged_two_colour_soliton_sqm_embedding_closed}=\text{false}, \\
&\quad \text{ap_e4_worldline_completion_closed}=\text{false}, \\
&\quad \text{route_e_portal_closed}=\text{false}, \\
&\text{six_dimensional_anomaly_gate_closed}=\text{false}, \\
&\quad \text{physics_promotion_allowed}=\text{false}.
\end{aligned} \tag{44}$$

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